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Designing a G-code file sharing platform

Semester: Spring 25-26

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1. PROJECT PROPOSAL

1.1 Background to the Problem

Nowadays, 3D printing and CNC machines are widely used in engineering projects, artwork, furniture design, and construction. These systems use 3D design files (like STL, OBJ) and G-code to create real objects. However, making a correct 3D model takes a lot of time and often needs many tries. G-code also needs to be adjusted for each design and machine, which can be difficult for many users.

At present, there is no single, well-organized platform that provides both high-quality 3D models and CNC G-code files along with proper guidance and support. Most existing platforms focus only on file sharing and downloading, but they lack strong quality control mechanisms, detailed instructions, and problem-solving features. Secondly there is no particular platform that provide CNC machine's file and suitable material, also existing platform don't provide file for different type of machine like (3 axis,2 axis). As a result, 3d designed material business has become complex
users often download files that are incomplete, poorly optimized, or incompatible with their machines, leading to failed prints, machining errors, and waste of time and materials additionally this every 3d designing

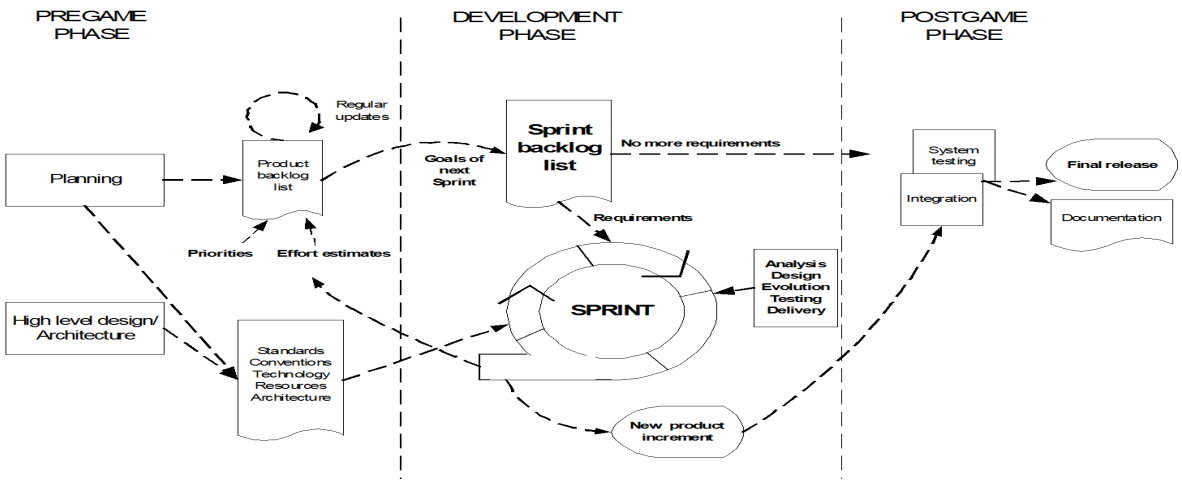
To solve this problem, this project proposes a platform called "GCodeHub". In this platform, users can upload, share, and review 3D models and G-code files. They can also discuss problems and get solutions from others. The system will use a credit-based method, where users earn credits by uploading files and use those credits to download files from others. It will also include an automatic system to check file quality and remove duplicate designs. In addition, users can hire designers for custom work.

The target users of this system are 3D printing users, CNC machine operators, students, hobbyists, and designers. These users can easily share their work and get useful files from others.

1.2 Selection of Process Model

For the GCodeHub project, we chose the Scrum-based Agile model. Scrum is a flexible way of developing software step by step with a team. It is suitable for this project because GCodeHub has many parts like user registration, file upload, file checking, duplicate detection, credit system, discussion forum, designer hiring, and admin control. These parts can be built one by one instead of building everything at the same time.

Scrum has three main phases: Pre-game, Game, and Post-game.



In the Pre-game phase, we plan everything. We collect requirements, make a list of tasks , decide priorities, and design the system. For GCodeHub, this includes defining features, writing user stories, designing the database, planning the system structure, and creating UI designs.

In the Game phase, we develop the system in small steps called Sprints. Each Sprint focuses on a specific feature. For example, one Sprint can build the login system, another can build file upload, and another can build the credit system. This makes the work easier and more organized.

In the Post-game phase, we prepare the final system. This includes combining all parts, testing the system, improving the design, writing documentation, and preparing the final report. This ensures everything works correctly.

Scrum is better than traditional models because requirements can change. For example, features like file checking, duplicate detection, or credit rules may need updates after feedback. Scrum allows us to make changes easily without starting over.

Scrum also helps in teamwork. The Product Owner manages tasks, the Scrum Master guides the process, and the team members build the system. This helps in dividing work, tracking progress, and testing regularly.

Although Extreme Programming (XP) is also an Agile method, it mainly focuses on coding practices. Scrum is better for this project because it helps more with planning, teamwork, and overall project management.

So, Scrum is chosen because it supports flexibility, step-by-step development, teamwork, and continuous improvement.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

2.1 Scopes and Features

Feature	Role	Scope
User Registration	Guest	Allows new users to create an account.
User Login	Guest / User / Designer / Admin	Allows users to access the system.
User Logout	User / Designer / Admin	Allows users to exit the system safely.
View Public Pages	Guest	Allows guests to view public pages.

Manage Profile	User / Designer / Admin	Allows users to view and update profile.
Change Password	User / Designer / Admin	Allows users to change account password.
View Files	Guest / User / Designer	Allows users to view available files.
Search Files	Guest / User / Designer	Allows users to search for files.
Download Files	Registered User	Allows users to download selected files.
Comment on File	Registered User	Allows users to comment on files.
Rate File	Registered User	Allows users to rate files.
Report File	Registered User	Allows users to report problematic files.
Designer Profile	Designer	Allows designers to manage profile details.
Hire Designer	Registered User	Allows users to send hire requests.
Manage Hire Requests	Designer	Allows designers to accept or reject requests.
Manage Users	Admin	Allows admin to manage user accounts.
Manage Files	Admin	Allows admin to manage uploaded files.
Manage Reports	Admin	Allows admin to review reported files.
Manage Comments	Admin	Allows admin to manage user comments.
Manage Ratings	Admin	Allows admin to monitor file ratings.
Manage Designers	Admin	Allows admin to manage designer profiles.
Monitor Activity Logs	Admin	Allows admin to view system activities.
Admin Dashboard	Admin	Allows admin to control system features.
Designer Dashboard	Designer	Allows designers to manage design-related tasks.
File Upload	Registered User	Allows users to upload files.

2.2 User Story Table

Item No	Specification [Role]	User Story	Acceptance Criteria
Feature: 1	**Landing & Public Access**		
1	Mission and vision [Guest]	As a guest, I want to understand the purpose of GCodeHub quickly so that I can decide whether to explore or register.	- Logo and navigation are visible - explain the platform - Buttons redirect to Browse Files and Sign Up - Feature are visible - Preview available file.C14 - Footer links are displayed
2	Public marketplace preview [Guest]	As a guest, I want to preview available files before creating an account.	- Guest can view file cards - Download button redirects to Login/Register - Search and category preview are visible - Restricted actions are protected
Feature: 2	**Authentication**		
4	Signup [Guest]	As a new visitor, I want to create an account using my personal information and selected role, so that I can access GCodeHub features such as uploading files, downloading files, earning credits, joining discussions, and requesting designer services.	- Clicking Sign Up will take the user to the Sign Up page. - Clicking Create Account with valid information will create a new account. - Clicking Create Account with empty required fields will show validation messages. - Entering an invalid email will show an email validation error.

			- Entering mismatched password and confirm password will show a mismatch error. - Selecting User role will create a regular user account. - Selecting Designer role will create a designer account. - Registering with an existing email will show an email already exists message. - Successful registration will take the user to the Login page or Dashboard page. - Clicking Login will take the user to the Login page. - Clicking the logo or Back to Home will take the user to the Landing page.
5	sign in [User, Designer, Admin]	As a registered user, I want to log in securely so that I can access my dashboard.	- Email must match a registered account - Password must match stored password - Invalid credentials show error message - Correct credentials redirect user according to role - Forgot password link is available
6	Password reset [User, Designer, Admin]	As a user, I want to reset my password if I forget it.	- Email must exist in the database - Reset code/message is sent - New password must follow password rules - Confirm password must match - Successful reset shows confirmation
Feature: 3	**Dashboard**		
7	Role-based navigation and logout [User, Designer, Admin]	As a logged-in user, I want clear navigation so that I can move between features easily.	- Sidebar shows role-appropriate menu items - Top bar shows search, credit balance, notifications, and profile menu - Logout clears session - Protected pages are not accessible after logout
8	System messages /Notification[User, Designer, Admin]	As a user, I want to receive important platform updates.	- Notifications are listed with date/time - Unread notifications are highlighted - Clicking a notification opens related page - Empty state appears when no notifications exist

Feature: 4	**Download file**		
9	Search and filter [User, Designer]	As a user, I want to browse G-code and 3D model files so that I can find useful files quickly.	- Search by title/keyword works - Filter by category, file type, machine type, material, credit cost, and rating works - File cards show title, thumbnail, type, rating, and required credits - No files found state is displayed when there are no results
10	Credit-based file download [User, Designer]	As a user, I want to download verified files using my credits.	- Only verified files can be downloaded - User must have enough credits - Required credits are deducted after successful download - Download transaction is stored - Download success/failure message is shown
Feature: 5	**Upload File**		
11	Upload 3D model or G-code file [User, Designer]	As a user, I want to upload files so that I can share my work and earn credits.	- Title, description, category, file type, machine type, material, and required credits are required - File drag-and-drop area accepts allowed file types only - Thumbnail upload accepts image files only - Submitted files move to Pending status - User sees validation progress
12	Detect duplicate or similar files during upload [System, Admin]	As the system, I need to check duplicate files so that repeated files are reduced.	- Uploaded file is checked against existing files - Duplicate status is stored - Duplicate warning is shown to admin - Duplicate file can be rejected or sent for manual review
13	Check file quality and compatibility [System, Admin]	As an admin, I want validation results so that only useful files are approved.	- Quality check result is visible - Unsupported or corrupted files are flagged - Admin can approve or reject - Rejection reason is required when rejecting
Feature: 6	**Credit Wallet**		
14	View current credits and summary [User, Designer]	As a user, I want to see my credit balance so that I know whether I can download files.	- Current balance is shown - Earned credits are shown - Spent credits are shown - Balance never becomes negative
15	Record credit earning and spending [User, Designer, Admin]	As a user, I want to view my credit transactions for transparency.	- Transaction ID, type, amount, description, and date are shown - Upload approval adds credits - Download deducts credits - Transactions are sorted by latest date

16	Check file quality and compatibility [System, Admin]	As an admin, I want validation results so that only useful files are approved.	- Quality check result is visible - Unsupported or corrupted files are flagged - Admin can approve or reject - Rejection reason is required when rejecting
Feature: 7	**Review**		
17	Add Review[User, Designer]	As a user, I want to review downloaded files so that others can judge file quality.	- Only users who downloaded the file can review - Rating must be between 1 and 5 - Comment field is optional but visible - Review appears on file details page
18	Manage Review [User, Designer]	As a user, I want to update my review if my opinion changes.	- User can edit only own review - User can delete only own review - Average rating updates after change - Admin can remove inappropriate reviews
Feature: 8	**Discussion**		
19	Discussion List[User, Designer]	As a user, I want to browse discussions so that I can solve CNC or 3D printing problems.	- Search discussion works - Category tabs are available - Post cards show title, author, comments count, tags, and status - Solved/Unsolved badge is visible
20	Discussion Details[User, Designer]	As a user, I want to read discussion details and comments.	- Post title, author, full content, and tags are visible - Comments are listed in order - Logged-in users can add comments - Post owner can mark discussion as solved
21	Create Discussion Post [User, Designer]	As a user, I want to ask questions when I face file or machine problems.	- Title and description are required - Category/tag selection is required - Submitted post appears in discussion list - Empty or invalid post is rejected
Feature: 9	**Profile Management**		
22	View Profile [User, Designer, Admin]	As a user, I want to view my profile and activity summary.	- Name, email, role, credit balance, uploaded files, and reviews are visible - No required field is empty - Edit profile button opens edit page
23	Edit Profile [User, Designer, Admin]	As a user, I want to update my profile information.	- Name cannot be empty - Email must be valid and unique - Profile image must be png/jpg - Save button updates information - Success/error message is shown
24	Change Password [User, Designer, Admin]	As a user, I want to change my password for security.	- Current password must match database password

			- New password must be at least 8 characters - Confirm password must match - Successful change shows confirmation
Feature: 10	**Admin Management**		
25	Admin Dashboard [Admin]	As an admin, I want to monitor users, files, reports, and pending approvals.	- Total users, total files, pending approvals, and reports are shown - Pending approval table is visible - Recent reports table is visible - Admin navigation works
26	Admin File Review [Admin]	As an admin, I want to review uploaded files before they become available.	- Pending files are listed - File metadata and validation result are visible - Approve changes status to Verified - Reject requires reason and changes status to Rejected - User is notified about decision
27	Admin User Management [Admin]	As an admin, I want to manage user status and roles.	- User table shows ID, name, email, role, credit balance, and status - Admin can view, suspend, or activate users - Suspended users cannot perform protected actions
28	Admin Reports [Admin]	As an admin, I want to manage reports so that harmful or duplicate content can be controlled.	- Report table shows report ID, reported item, reason, submitted by, status, and date - Admin can view report details - Admin can resolve or dismiss report - Status updates after action
Feature: 11	**Designer Dashboard**		
29	Designer Dashboard [Designer]	As a designer, I want to view and manage my custom design work requests from a dashboard, so that I can track incoming requests, accepted work, completed work, ratings, and earned credits in one place.	- Clicking Designer Dashboard will take the designer to the Designer Dashboard page. - Designer Dashboard will show total active requests. - Designer Dashboard will show total completed work. - Designer Dashboard will show designer rating. - Designer Dashboard will show earned credits. - Designer Dashboard will show incoming custom work requests. - Clicking View Request will take the designer to the request details page. - Clicking Accept Request will change the request status to Accepted.

			- Clicking Reject Request will change the request status to Rejected. - Clicking Active Work will show the list of currently accepted work. - Clicking Submit Work will take the designer to the submit completed work page. - Clicking Portfolio will take the designer to the portfolio management section/page. - Clicking Update Availability will allow the designer to change availability status. - If there are no incoming requests, the system will show “No new custom requests available.”
Feature: 12	**Hire Designer**		
30	Hire Designer [Designer,User]	As a registered user, I want to browse available designers and hire a suitable designer for custom G-code or 3D model work, so that I can get a machine-ready file based on my requirements.	- Clicking Hire Designer will take the user to the Hire Designer page. - Hire Designer page will show a list of available designers. - Each designer card will show designer name, skills, rating, availability, and profile summary. - Clicking Search will filter designers based on entered keyword. - Clicking skill filter will show designers matching the selected skill. - Clicking rating filter will show designers based on selected rating range. - Clicking availability filter will show available designers only. - Clicking View Profile will take the user to the selected designer profile page. - Clicking Hire will take the user to the Custom Work Request page. - User can enter custom work title, description, budget credits, deadline, and reference file. - Clicking Submit Request with valid information will create a custom work request. - Clicking Submit Request with empty required fields will show validation messages. - After successful request submission, the system will show

			"Custom work request submitted successfully." - If no designer matches the filter, the system will show "No designers found."

2.3 Functional Requirements

Item No	Feature	Specification	Screen Definition	User Story	Acceptance Criteria
1	Landing & Public Access	Mission and vision [Guest]	Landing page with logo, navigation, platform explanation, call-to-action buttons, file preview, and footer.	As a guest, I want to understand the purpose of GCodeHub quickly so that I can decide whether to explore or register.	- Logo and navigation are visible - explain the platform - Buttons redirect to Browse Files and Sign Up - Feature are visible - Preview available file. - Footer links are displayed
2	Landing & Public Access	Public marketplace preview [Guest]	Public marketplace preview with file cards, search/category preview	As a guest, I want to preview available files before creating an account.	- Guest can view file cards - Download button redirects to Login/Register - Search and category preview are visible - Restricted actions are protected
4	Authentication	Signup [Guest]	Sign Up page with personal information form, role selection, validation messages, and navigation to login/home.	As a new visitor, I want to create an account using my personal information and selected role, so that I can access GCodeHub features such as uploading files, downloading files, earning credits, joining	- Clicking Sign Up will take the user to the Sign Up page. - Clicking Create Account with valid information will create a new account. - Clicking Create Account with empty required fields will show validation messages. - Entering an invalid email will show an email validation error. - Entering mismatched password and confirm password will show a mismatch error.

				discussions, and requesting designer services.	- Selecting User role will create a regular user account. - Selecting Designer role will create a designer account. - Registering with an existing email will show an email already exists message. - Successful registration will take the user to the Login page or Dashboard page. - Clicking Login will take the user to the Login page. - Clicking the logo or Back to Home will take the user to the Landing page.
5	Authentication	sign in [User, Designer, Admin]	Login page with email, password, forgot password link, error message, and role-based redirection.	As a registered user, I want to log in securely so that I can access my dashboard.	- Email must match a registered account - Password must match stored password - Invalid credentials show error message - Correct credentials redirect user according to role - Forgot password link is available
6	Authentication	Password reset [User, Designer, Admin]	Password reset screen with email verification, reset code/message, new password, confirm password, and confirmation.	As a user, I want to reset my password if I forget it.	- Email must exist in the database - Reset code/message is sent - New password must follow password rules - Confirm password must match - Successful reset shows confirmation
7	Dashboard	Role-based navigation and logout [User, Designer, Admin]	Dashboard layout with role-based sidebar, top bar, search, credit balance, notifications, profile menu, and logout.	As a logged-in user, I want clear navigation so that I can move between features easily.	- Sidebar shows role-appropriate menu items - Top bar shows search, credit balance, notifications, and profile menu - Logout clears session - Protected pages are not accessible after logout
8	Dashboard	System messages /Notification [User, Designer, Admin]	Notification page or panel with notification list, unread highlight, related page	As a user, I want to receive important platform updates.	- Notifications are listed with date/time - Unread notifications are highlighted - Clicking a notification opens related page - Empty state appears when no notifications exist

			link, and empty state.		
9	Download file	Search and filter [User, Designer]	Browse files screen with search bar, filters, file cards, file metadata, rating, required credits, and no-result state.	As a user, I want to browse G-code and 3D model files so that I can find useful files quickly.	- Search by title/keyword works - Filter by category, file type, machine type, material, credit cost, and rating works - File cards show title, thumbnail, type, rating, and required credits - No files found state is displayed when there are no results
10	Download file	Credit-based file download [User, Designer]	File detail/download screen with verified file status, required credit check, transaction saving, and success/failure message.	As a user, I want to download verified files using my credits.	- Only verified files can be downloaded - User must have enough credits - Required credits are deducted after successful download - Download transaction is stored - Download success/failure message is shown
11	Upload File	Upload 3D model or G-code file [User, Designer]	Upload file screen with metadata form, drag-and-drop file upload, thumbnail upload, and validation/progress display.	As a user, I want to upload files so that I can share my work and earn credits.	- Title, description, category, file type, machine type, material, and required credits are required - File drag-and-drop area accepts allowed file types only - Thumbnail upload accepts image files only - Submitted files move to Pending status - User sees validation progress
12	Upload File	Detect duplicate or similar files during upload [System, Admin]	Duplicate check result screen for system/admin with duplicate warning, stored duplicate status, and manual review action.	As the system, I need to check duplicate files so that repeated files are reduced.	- Uploaded file is checked against existing files - Duplicate status is stored - Duplicate warning is shown to admin - Duplicate file can be rejected or sent for manual review
13	Upload File	Check file quality and compatibilit	Admin file validation screen	As an admin, I want validation results so that only	- Quality check result is visible - Unsupported or corrupted files are flagged

		y [System, Admin]	showing quality check result, unsupported/corrupted file flag, approve/reject option, and rejection reason.	useful files are approved.	- Admin can approve or reject - Rejection reason is required when rejecting
14	Credit Wallet	View current credits and summary [User, Designer]	Credit wallet screen showing current balance, earned credits, spent credits, and balance summary.	As a user, I want to see my credit balance so that I know whether I can download files.	- Current balance is shown - Earned credits are shown - Spent credits are shown - Balance never becomes negative
15	Credit Wallet	Record credit earning and spending [User, Designer, Admin]	Credit transaction history screen with transaction ID, type, amount, description, date, and latest-first sorting.	As a user, I want to view my credit transactions for transparency.	- Transaction ID, type, amount, description, and date are shown - Upload approval adds credits - Download deducts credits - Transactions are sorted by latest date
16	Credit Wallet	Check file quality and compatibility [System, Admin]	Admin validation screen showing quality result, unsupported/corrupted file flag, approve/reject option, and rejection reason.	As an admin, I want validation results so that only useful files are approved.	- Quality check result is visible - Unsupported or corrupted files are flagged - Admin can approve or reject - Rejection reason is required when rejecting
17	Review	Add Review[User, Designer]	File review section with rating input, optional comment field, downloaded-file permission check, and	As a user, I want to review downloaded files so that others can judge file quality.	- Only users who downloaded the file can review - Rating must be between 1 and 5 - Comment field is optional but visible - Review appears on file details page

			review display.		
18	Review	Manage Review [User, Designer]	Manage review screen with user-owned review edit/delete options, average rating update, and admin moderation support.	As a user, I want to update my review if my opinion changes.	- User can edit only own review - User can delete only own review - Average rating updates after change - Admin can remove inappropriate reviews
19	Discussion	Discussion List[User, Designer]	Discussion list screen with search, category tabs, post cards, author, comments count, tags, and solved/unsolved badge.	As a user, I want to browse discussions so that I can solve CNC or 3D printing problems.	- Search discussion works - Category tabs are available - Post cards show title, author, comments count, tags, and status - Solved/Unsolved badge is visible
20	Discussion	Discussion Details[User, Designer]	Discussion details screen with post title, author, full content, tags, ordered comments, add comment, and mark as solved option.	As a user, I want to read discussion details and comments.	- Post title, author, full content, and tags are visible - Comments are listed in order - Logged-in users can add comments - Post owner can mark discussion as solved
21	Discussion	Create Discussion Post [User, Designer]	Create discussion post screen with title, description, category/tag selection, validation, and submission result.	As a user, I want to ask questions when I face file or machine problems.	- Title and description are required - Category/tag selection is required - Submitted post appears in discussion list - Empty or invalid post is rejected
22	Profile Management	View Profile [User, Designer, Admin]	Profile page showing name, email, role, credit balance, uploaded files, reviews, activity	As a user, I want to view my profile and activity summary.	- Name, email, role, credit balance, uploaded files, and reviews are visible - No required field is empty - Edit profile button opens edit page

			summary, and edit button.		
23	Profile Management	Edit Profile [User, Designer, Admin]	Edit profile screen with name, email, profile image upload, save button, validation, and success/error message.	As a user, I want to update my profile information.	- Name cannot be empty - Email must be valid and unique - Profile image must be png/jpg - Save button updates information - Success/error message is shown
24	Profile Management	Change Password [User, Designer, Admin]	Change password screen with current password, new password, confirm password, validation, and confirmation message.	As a user, I want to change my password for security.	- Current password must match database password - New password must be at least 8 characters - Confirm password must match - Successful change shows confirmation
25	Admin Management	Admin Dashboard [Admin]	Admin dashboard with summary cards, pending approvals table, recent reports table, and admin navigation.	As an admin, I want to monitor users, files, reports, and pending approvals.	- Total users, total files, pending approvals, and reports are shown - Pending approval table is visible - Recent reports table is visible - Admin navigation works
26	Admin Management	Admin File Review [Admin]	Admin file review screen with pending file list, metadata, validation result, approve/reject actions, and user notification result.	As an admin, I want to review uploaded files before they become available.	- Pending files are listed - File metadata and validation result are visible - Approve changes status to Verified - Reject requires reason and changes status to Rejected - User is notified about decision
27	Admin Management	Admin User Managemen t [Admin]	Admin user management screen with user table, role, credit balance, status, and	As an admin, I want to manage user status and roles.	- User table shows ID, name, email, role, credit balance, and status - Admin can view, suspend, or activate users - Suspended users cannot perform protected actions

			activate/suspe nd actions.		
28	Admin Management	Admin Reports [Admin]	Admin reports screen with report table, report details, resolve/dismis s actions, status update, and date.	As an admin, I want to manage reports so that harmful or duplicate content can be controlled.	- Report table shows report ID, reported item, reason, submitted by, status, and date - Admin can view report details - Admin can resolve or dismiss report - Status updates after action
29	Designer Dashboard	Designer Dashboard [Designer]	Designer dashboard with active requests, completed work, rating, earned credits, incoming requests, active work, portfolio, and availability controls.	As a designer, I want to view and manage my custom design work requests from a dashboard, so that I can track incoming requests, accepted work, completed work, ratings, and earned credits in one place.	- Clicking Designer Dashboard will take the designer to the Designer Dashboard page. - Designer Dashboard will show total active requests. - Designer Dashboard will show total completed work. - Designer Dashboard will show designer rating. - Designer Dashboard will show earned credits. - Designer Dashboard will show incoming custom work requests. - Clicking View Request will take the designer to the request details page. - Clicking Accept Request will change the request status to Accepted. - Clicking Reject Request will change the request status to Rejected. - Clicking Active Work will show the list of currently accepted work. - Clicking Submit Work will take the designer to the submit completed work page. - Clicking Portfolio will take the designer to the portfolio management section/page. - Clicking Update Availability will allow the designer to change availability status. - If there are no incoming requests, the system will show “No new custom requests available.”
30	Hire Designer	Hire Designer	Hire designer screen with designer list,	As a registered user, I want to browse available	- Clicking Hire Designer will take the user to the Hire Designer page.

		[Designer,User]	search/filter options, profile view, hire action, custom request form, and submission result.	designers and hire a suitable designer for custom G-code or 3D model work, so that I can get a machine-ready file based on my requirements.	- Hire Designer page will show a list of available designers. - Each designer card will show designer name, skills, rating, availability, and profile summary. - Clicking Search will filter designers based on entered keyword. - Clicking skill filter will show designers matching the selected skill. - Clicking rating filter will show designers based on selected rating range. - Clicking availability filter will show available designers only. - Clicking View Profile will take the user to the selected designer profile page. - Clicking Hire will take the user to the Custom Work Request page. - User can enter custom work title, description, budget credits, deadline, and reference file. - Clicking Submit Request with valid information will create a custom work request. - Clicking Submit Request with empty required fields will show validation messages. - After successful request submission, the system will show “Custom work request submitted successfully.” - If no designer matches the filter, the system will show “No designers found.”

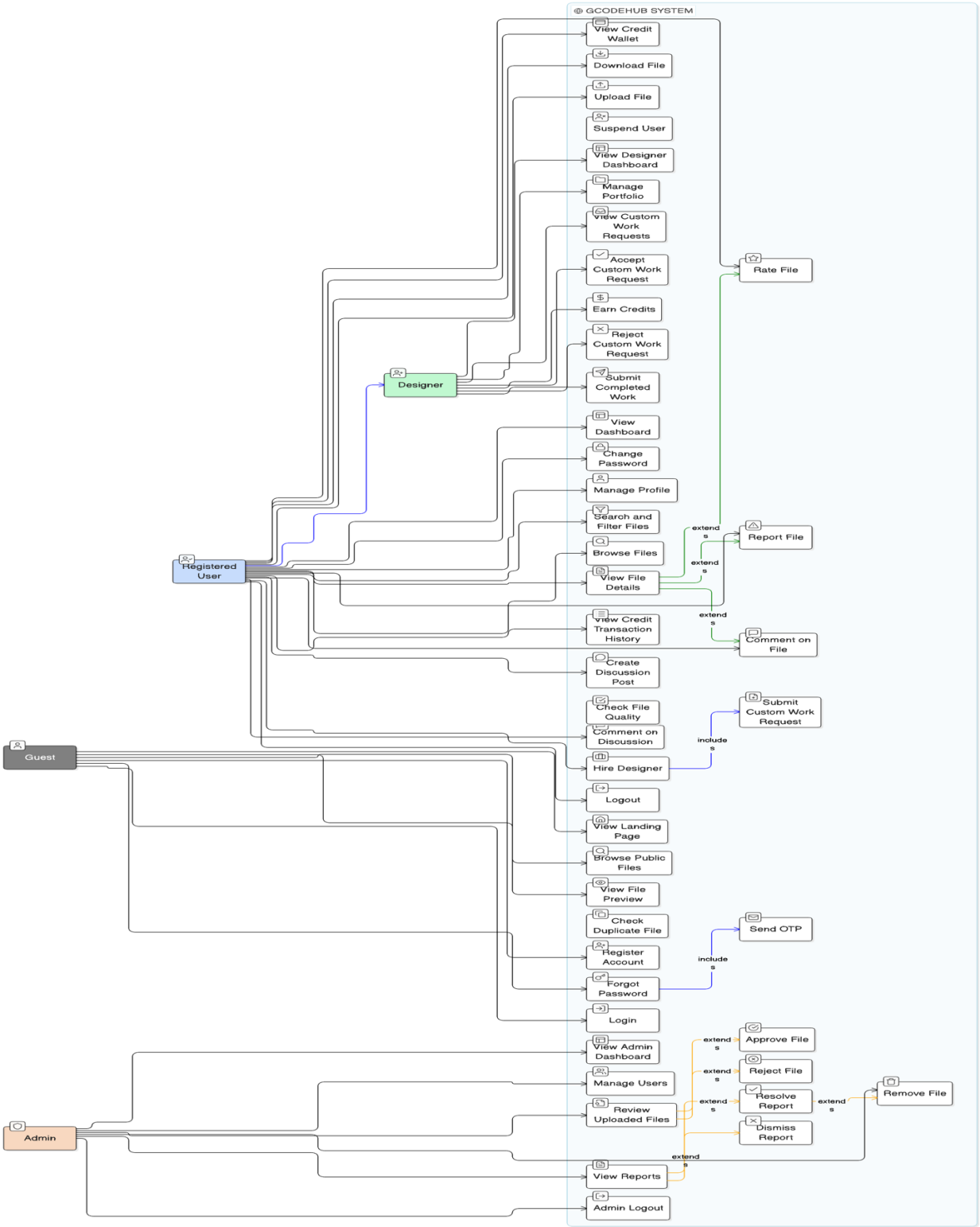
2.3.2 Non-Functional Requirements

Item No	Requirement Type	Requirement	Acceptance Criteria
1	Security	The system shall protect user accounts, file access, credits, and admin features from unauthorized access.	Only authenticated and authorized roles can access protected pages and actions.
2	Authentication and Authorization	The system shall allow role-based access for Guest, User, Designer, and Admin.	Users are redirected or blocked according to their assigned role.
3	Password Protection	The system shall store passwords securely using password hashing instead of plain text.	No plain text password is stored or displayed in the database or interface.
4	Performance	The system shall load common pages and return search/filter results within a reasonable time under normal usage.	Landing, dashboard, file listing, and search results load without noticeable delay.
5	Reliability	The system shall handle downloads, credit deductions, uploads, and request submissions consistently.	Credits are deducted only after successful download and duplicate transactions are prevented.
6	Usability	The system shall provide clear navigation, readable pages, useful messages, and simple user flows.	Users can register, log in, browse files, download files, and hire designers without confusion.
7	Availability	The system shall remain available to users during normal operating time except maintenance periods.	Users can access public pages and protected dashboards when the system is online.
8	Compatibility	The web application shall work on common modern browsers and responsive screen sizes.	Core pages work properly on Chrome, Firefox, Edge, desktop, and mobile-size screens.
9	Maintainability	The system shall be organized into clear modules such as authentication, files, wallet, discussions, designer, and admin.	Future feature updates and bug fixes can be performed without changing unrelated modules.

3. SOFTWARE DESIGN

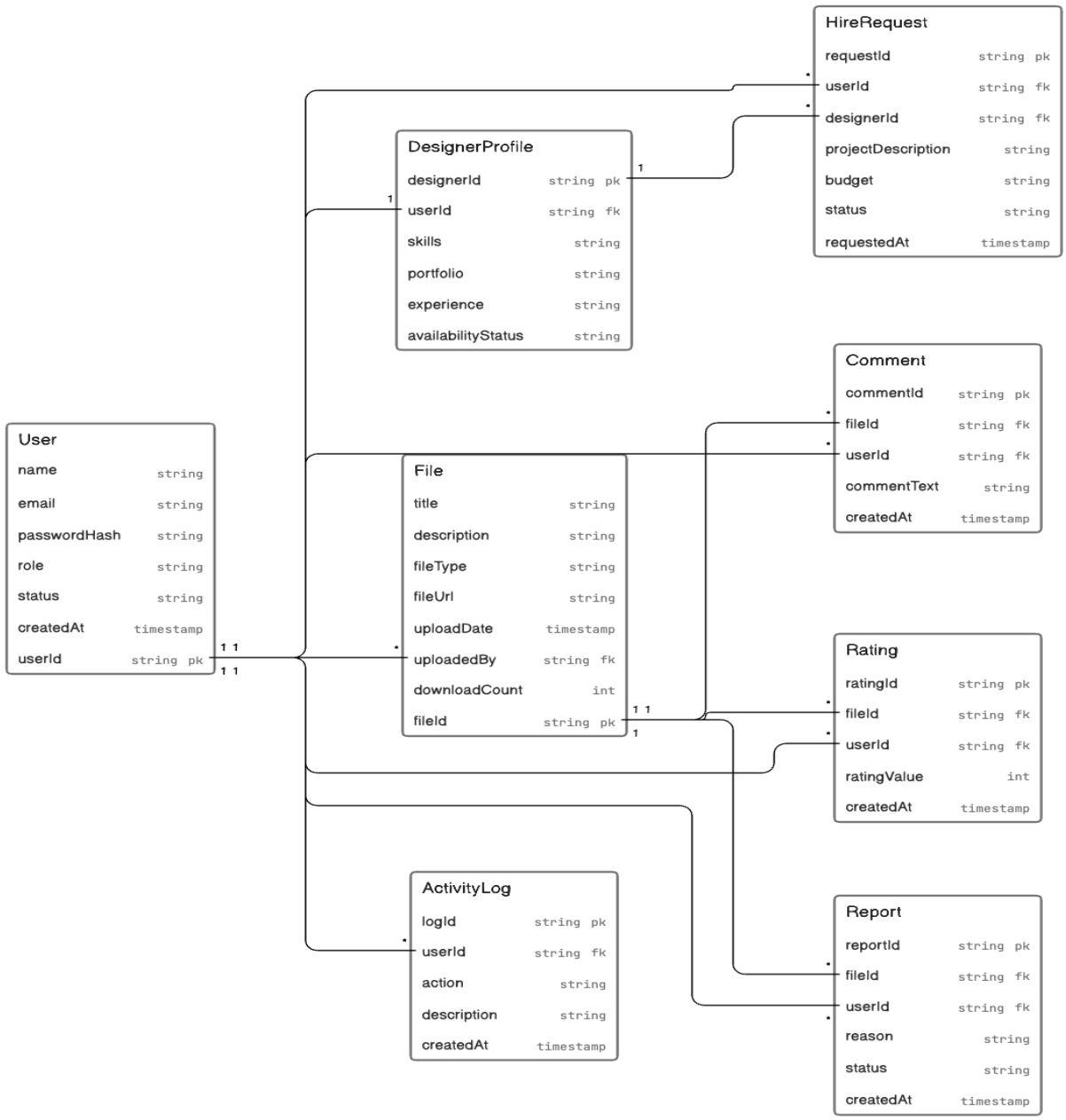
3.1 System Design:

3.1.1 Use Case Diagram



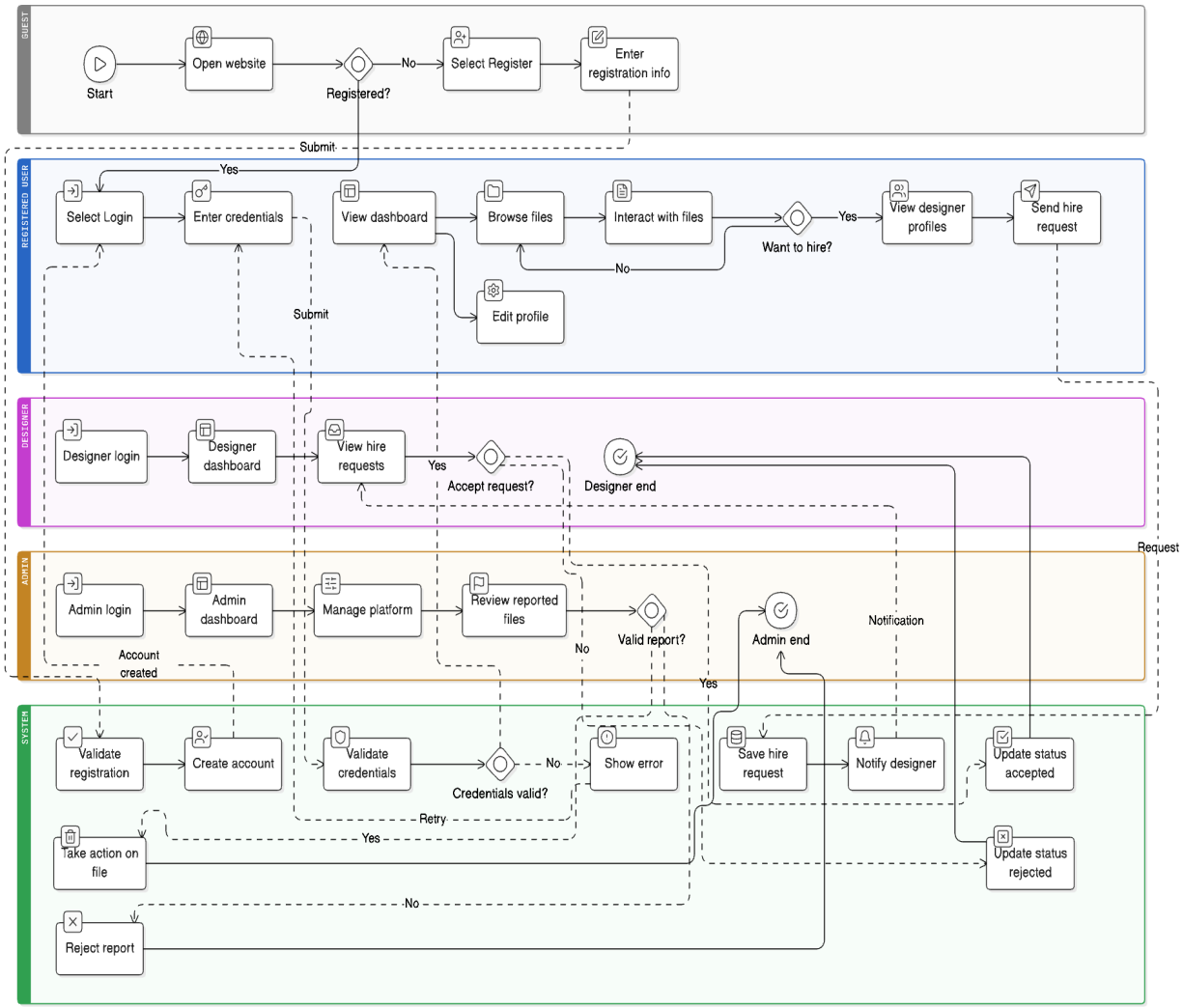
The **Use Case Diagram** shows how different actors interact with the Digital File Sharing and Designer Hiring Platform. The main actors are **Guest, Registered User, Designer, and Admin**. A guest can register, log in, and view public pages. A registered user can view files, download files, comment, rate, report files, manage profile, and hire designers. A designer can manage their designer profile and respond to hire requests. The admin can manage users, files, reports, comments, ratings, designers, and monitor system activity.

3.1.2 Class Diagram



The **Class Diagram** shows the main classes, attributes, methods, and relationships of the system. The major classes are **User, DesignerProfile, File, Comment, Rating, Report, HireRequest**. A user can upload files, comment on files, rate files, report files, send hire requests, and generate activity logs. A user may also have a designer profile. Files can have many comments, ratings, and reports. Hire requests connect registered users with designers. The admin is represented through the role attribute of the User class.

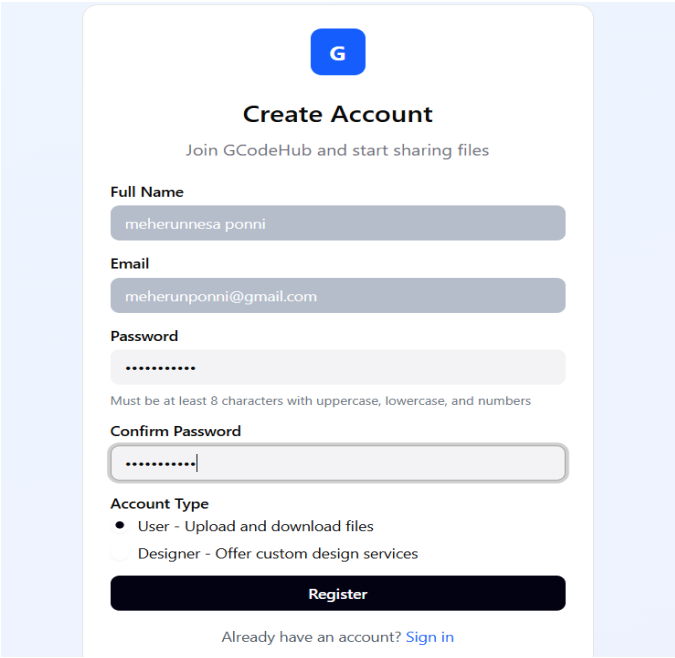
Activity Diagram



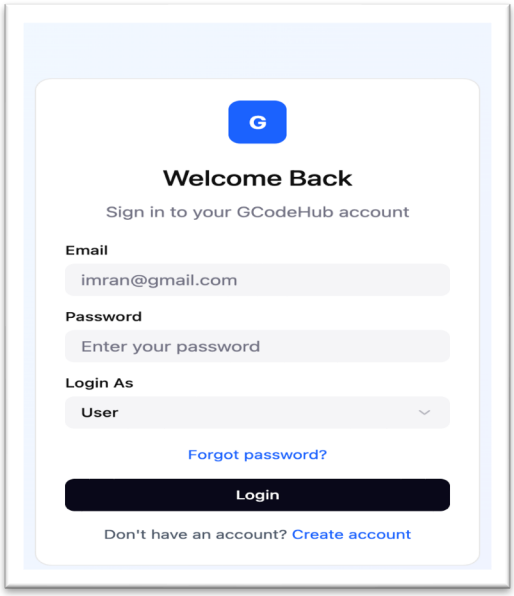
The **Activity Diagram** shows the workflow of the system from start to end. The user opens the website and either registers or logs in. After successful login, the user can access the dashboard, view or download files, comment, rate, report files, edit profile, and send hire requests to designers. Designers can accept or reject hire requests. Admins can log in to manage users, files, reports, comments, ratings, designers, and activity logs. The diagram represents the step-by-step flow of actions and decisions in the system.

3.2 UI / Wireframe Design using Figma

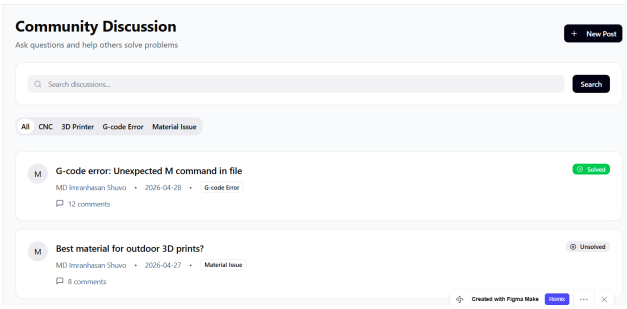
Signup Page



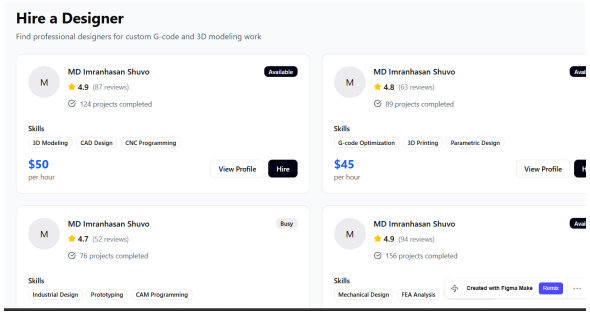
Login Page



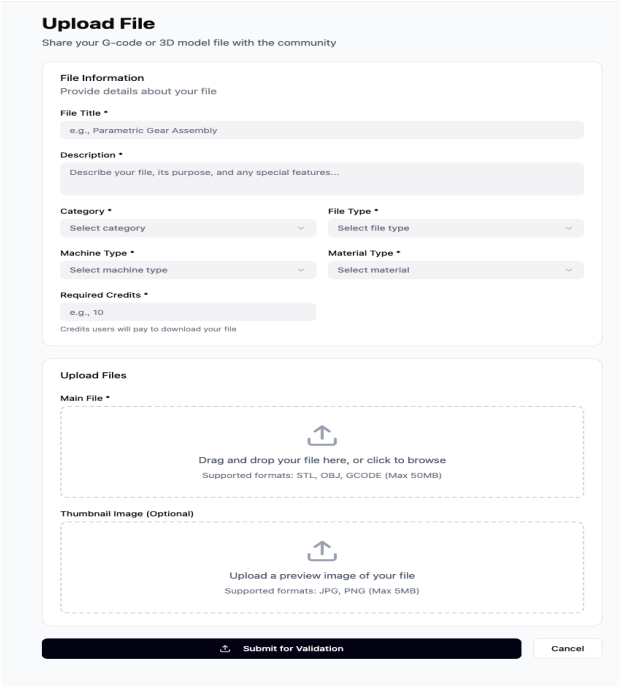
Community Discussion Page



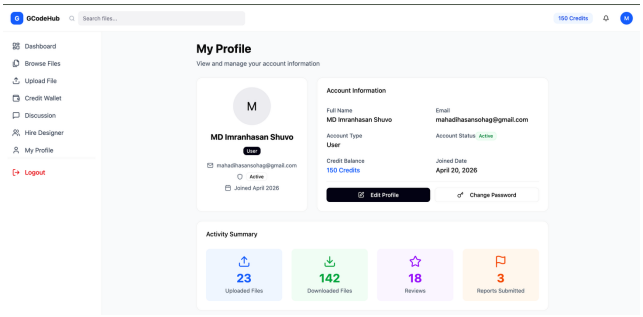
Hiring Page



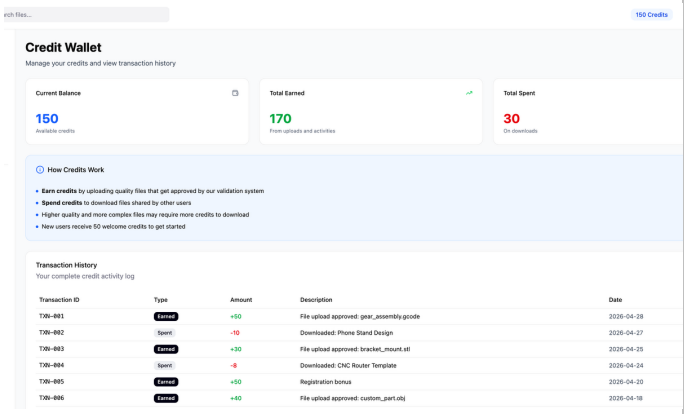
Upload File Page



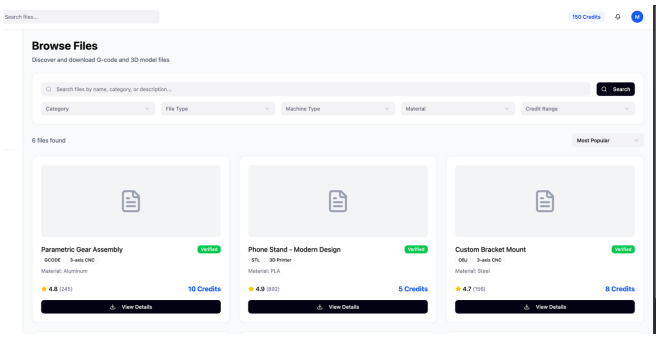
My Profile Page



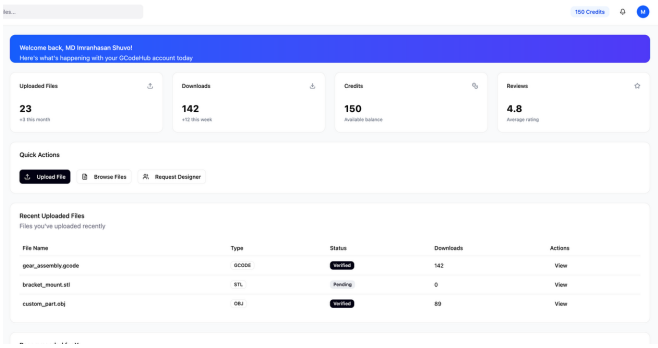
Credit Wallet Page



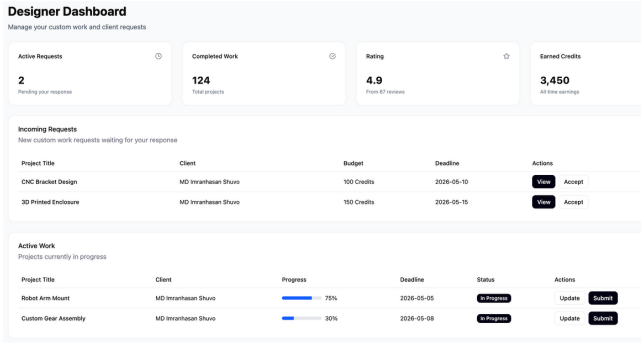
File Browsing Page



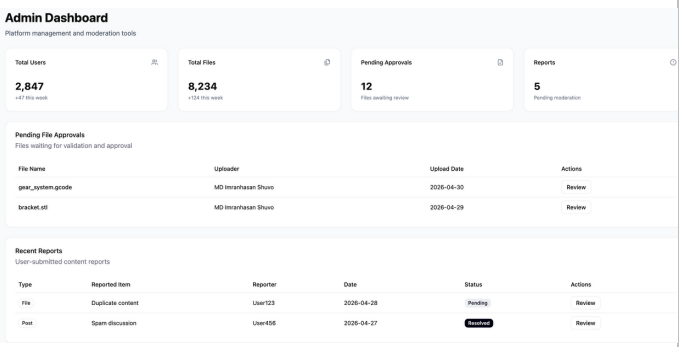
User Dashbord



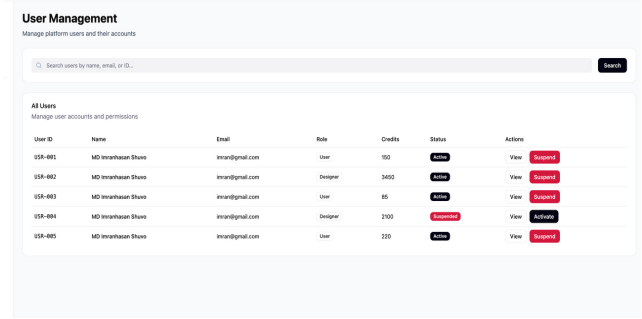
Designer Dashbord



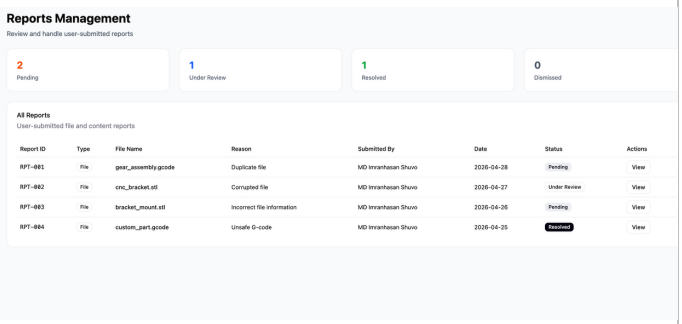
Admin Dashbord



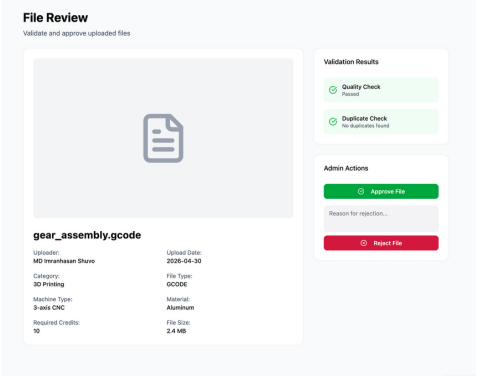
User Management Page



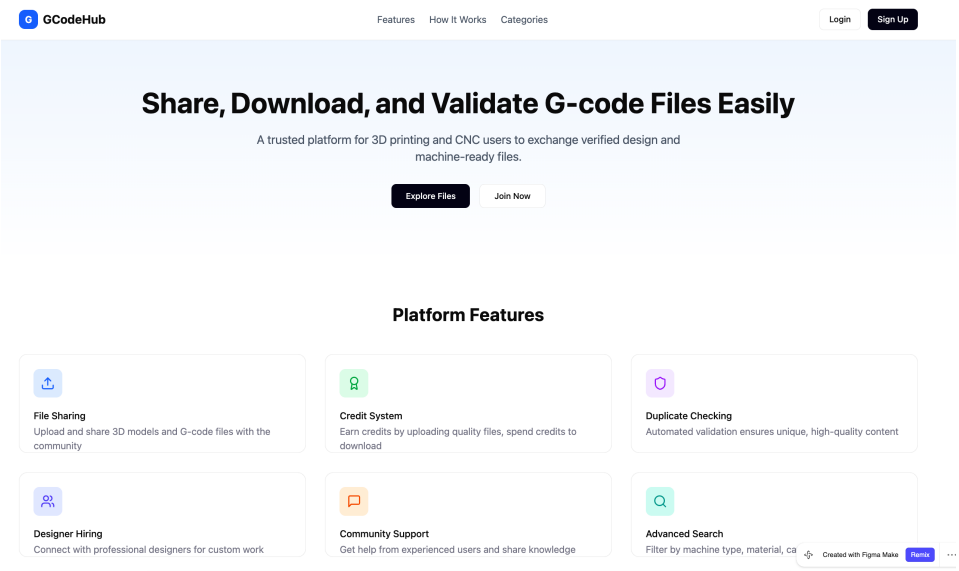
Report Management



File Review Page



Landing page



4. SOFTWARE TESTING

Project Name: GCodeHub		Test Designed by: Md Imran Hasan Shuvo		
Test Case ID: TC-01		Test Designed date: 30/04/2026		
Test Priority (Low, Medium, High): High		Test Executed by: Md Imran Hasan Shuvo		
Module Name: Authentication		Test Execution date: 30/04/2026		
Test Title: Verify user registration with valid information				
Description: Test that a user can register successfully using valid information				
Precondition: A guest user is on the signup page				
Dependencies: Valid name, email, strong password, user role				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open signup page 2. Enter valid name and email Enter password and confirm password 3. Select user role 4. Click Sign Up	Valid name, email, strong password, user role	Account should be created and user should be redirected to login/dashboard	As expecte d	Pass

Project Name: GCodeHub		Test Designed by: Md Imran Hasan Shuvo		
Test Case ID: TC-02		Test Designed date: 30/04/2026		
Test Priority (Low, Medium, High): High		Test Executed by: Md Imran Hasan Shuvo		
Module Name: Authentication		Test Execution date: 01/05/2026		
Test Title: Verify login with invalid password				
Description: Test that login is denied when an incorrect password is entered and an appropriate error message is displayed				
Precondition: User account exists				
Dependencies: Email: existing user, Password: wrong password				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open login page 2. Enter registered email 3. Enter wrong password 4. Click Login	Email: existing user, Password: wrong password	System should show invalid email or password message and should not open dashboard	As expected	Pass

Project Name: GCodeHub		Test Designed by: Md. Arafat Al Shahriar		
Test Case ID: TC-03		Test Designed date: 30/04/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: Md. Arafat Al Shahriar		
Module Name: Browse/Search		Test Execution date: 02/05/2026		
Test Title: Verify file search by machine type and material				
Description: Verify that users can search for files by selecting machine type and material, and that the correct results are displayed				
Precondition: There are published files in the database				
Dependencies: Machine type: 3-axis CNC, Material: Wood				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open browse page 2. Select machine type filter 3. Select material filter 4. Click Search	Machine type: 3-axis CNC, Material: Wood	Only matching files should be displayed	As expecte d	Pass

Project Name: GCodeHub		Test Designed by: Md. Arafat Al Shahriar		
Test Case ID: TC-04		Test Designed date: 30/04/2026		
Test Priority (Low, Medium, High): High		Test Executed by: Md. Arafat Al Shahriar		
Module Name: File Upload		Test Execution date: 02/05/2026		
Test Title: Verify upload with supported G-code file				
Description: Test that a user can successfully upload a supported G-code file				
Precondition: User is logged in				
Dependencies: File: sample.nc, Title, Description, Material, Machine Type				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open upload page 2. Fill all required metadata 3. Attach supported file 4. Click Submit	File: sample.nc, Title, Description, Material, Machine Type	File should be submitted for validation/admin approval	As expected	Pass

Project Name: GCodeHub		Test Designed by: Meherunnesa Ponni		
Test Case ID: TC-05		Test Designed date: 30/04/2026		
Test Priority (Low, Medium, High): High		Test Executed by: Meherunnesa Ponni		
Module Name: File Upload		Test Execution date: 03/05/2026		
Test Title: Verify upload rejection for unsupported file type				
Description: Verify that the system rejects uploads of unsupported file types and displays an appropriate error message				
Precondition: User is logged in				
Dependencies: File: document.pdf				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open upload page 2. Fill required metadata 3. Attach unsupported file 4. Click Submit	File: document.pdf	System should reject file and show unsupported format message	As expected	Pass

Project Name: GCodeHub		Test Designed by: Meherunnesa Ponni		
Test Case ID: TC-06		Test Designed date: 30/04/2026		
Test Priority (Low, Medium, High): High		Test Executed by: Meherunnesa Ponni		
Module Name: File Validation/Admin		Test Execution date: 03/05/2026		
Test Title: Verify admin approval of pending file				
Description: Test that an admin can approve a pending file and that its status is updated accordingly				
Precondition: Admin is logged in and pending file exists				
Dependencies: Pending file ID				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open admin panel 2. Open File Review 3. Select a pending file 4. Click Approve		File status should change to approved and become visible in browse page	As expected	Pass

Project Name: GCodeHub		Test Designed by: Tanjim Islam Zinnia		
Test Case ID: TC-07		Test Designed date: 30/04/2026		
Test Priority (Low, Medium, High): High		Test Executed by: Tanjim Islam Zinnia		
Module Name: Credit & Download		Test Execution date: 04/05/2026		
Test Title: Verify download when user has enough credits				
Description: Test that a user can successfully download a file when they have sufficient credits, and the credits are deducted correctly				
Precondition: User is logged in and has sufficient credits				
Dependencies: File cost: 10 credits, User credits: 130				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open file details page 2. Click Download 3. Confirm credit deduction	File cost: 10 credits, User credits: 130	File should download, credits should be deducted, and history should be saved	As expected	Pass. .

Project Name: GCodeHub		Test Designed by: Tanjim Islam Zinnia		
Test Case ID: TC-08		Test Designed date: 30/04/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: Tanjim Islam Zinnia		
Module Name: Review & Rating		Test Execution date: 04/05/2026		
Test Title: Verify review submission after download				
Description: Test that a user can submit a review after downloading a file and that the review is saved successfully				
Precondition: User has downloaded the file				
Dependencies: Rating: 4, Comment: Useful file				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open downloaded file details 2. Enter rating and comment 3. Click Submit Review	Rating: 4, Comment: Useful file	Review should be saved and shown under file details	As expected	Pass .

Project Name: GCodeHub		Test Designed by: MD. Fahid Hossain Anoy		
Test Case ID: TC-09		Test Designed date: 30/04/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: MD. Fahid Hossain Anoy		
Module Name: Discussion		Test Execution date: 05/05/2026		
Test Title: Verify discussion post creation under a file				
Description: Verify that a user can create a comment under a file and that it is displayed correctly				
Precondition: User is logged in				
Dependencies: Question: Which material is suitable?				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open file details page 2. Type question in comment box 3. Click Post	Question: Which material is suitable?	Question should appear in comment list	As expected	Pass.

Project Name: GCodeHub		Test Designed by: MD. Fahid Hossain Anoy		
Test Case ID: TC-10		Test Designed date: 30/04/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: MD. Fahid Hossain Anoy		
Module Name: Designer Hiring		Test Execution date: 05/05/2026		
Test Title: Verify custom design request submission				
Description: Test that a user can successfully submit a custom design request with valid details				
Precondition: User is logged in				
Dependencies: Requirement, budget, deadline				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open hire designer page 2. Fill requirement, budget, deadline,others 3. Click Submit Request	Title, budget, deadline,others	Hire request should be created and visible to designers	As expected	Pass.

5. CONCLUSION

GCodeHub is proposed as a structured G-code and 3D model file sharing platform that addresses a practical problem faced by 3D printing users, CNC machine operators, designers, students, and hobbyists. The project focuses on reducing the difficulty of finding reliable machine-ready files by combining file upload, validation, duplicate checking, reviews, discussion, credit-based exchange, designer hiring, and admin moderation in one system. The selected Agile Iterative Process Model is suitable because the project contains several connected modules that can be developed and tested sprint by sprint. The SRS, user stories, traceability matrix, system design diagrams, UI mapping, and manual test cases provide a complete foundation for implementation. If developed properly, the system can reduce failed printing or machining attempts, improve file quality, reward useful contributors, and create a helpful technical community. Future improvements may include advanced simulation, online payment for credits, AI-assisted file checking, and direct machine compatibility recommendations. Overall, GCodeHub is feasible, useful, and appropriate for a Software Engineering project because it combines real-world need with clear requirements and testable functionality.